A. WEKA EXPLORER

1. Which file formats does WEKA support?
2. what is the ARFF format and what form does it take?

Answer:

1. WEKA (Waikato Environment for Knowledge Analysis) is a popular open-source suite of machine learning software written in Java. It supports several file formats for loading datasets, each with its own structure and use cases:

a) ARFF (Attribute-Relation File Format): Native format for WEKA. Plain text format designed specifically for WEKA. Contains data descriptions and data instances.
b) CSV (Comma-Separated Values): Widely used for tabular data. Each row represents an instance, columns are attributes. May require conversion or additional specification for attribute types in WEKA.
c) C4.5 (used for decision trees): Includes .names and .data files. Used with C4.5/J48 decision tree algorithms.
d) XRFF (XML-based ARFF): XML version of ARFF. Useful for data interchange or applications needing XML.
e) JSON: Recent WEKA versions have support for JSON-formatted datasets.
f) LibSVM: Format used by Support Vector Machine libraries. Sparse representation of data.
g) Other Formats: WEKA can also import data from database sources (using JDBC). Some plugins and extensions provide support for additional formats (e.g., Excel .xls/.xlsx, HDF5).

2. What Is the ARFF Format and What Form Does It Take?

a) ARFF Format Overview
ARFF stands for Attribute-Relation File Format. It is a text file format developed specifically for WEKA to describe datasets. An ARFF file contains two main sections:
    - Header: Metadata about attributes (columns) and their types.
    - Data: Actual data instances (rows).
b) Structure of an ARFF File
    i. Header Section:
        - Begins with the relation name.
        - Lists each attribute's name and type.
        - Types can be numeric, nominal (enumerated), string, date, etc.
    ii. Data Section:
        - Begins with @data
        - Each line after represents one instance (row) of data.

c) Supported Attribute Types in ARFF

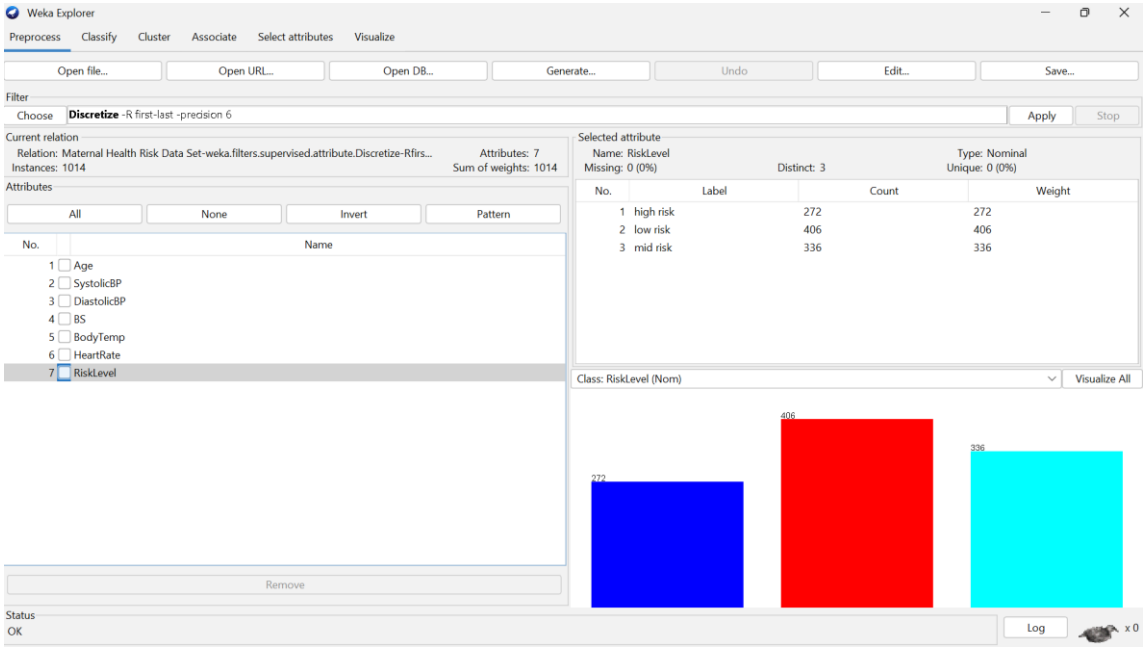
- Numeric: real or numeric (integers or floats).
- Nominal: Specific set of possible values.
- String: Arbitrary text strings.
- Date: Date/time values, usually with a format specification.

d) Comments

- Lines starting with % are comments and ignored by WEKA.

B. Data Set for Analysis

1. Search the repositories to select a dataset for analysis
   - UC Irvine Machine Learning Repository -https://archive.ics.uci.edu/datasets
   - Kaggle Repository-https://www.kaggle.com/datasets?fileType=csv
2. Prepare data for analysis, if necessary
   - discrete values of attributes
   - no missing values

Answer:

1. I have a dataset on https://archive-beta.ics.uci.edu/dataset/863/maternal+health+risk
2. We can see instances of 1014, attributes of 7, Nominal Attributes type, and no missing values